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Short communication

A TaqMan RT-qPCR assay for tilapia lake virus (TiLV) detection in tilapia

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ABSTRACT

Tilapia lake virus disease (TiLVD) caused by tilapia lake virus is a highly contagious disease affecting tilapia and its hybrid species. Thus far, the virus has been identified in > 10 countries across 3 continents. For this reason, reliable and rapid diagnostic assays are urgently required. Here, we describe the development and validation of a TaqMan probe-based reverse transcription quantitative polymerase chain reaction (RT-qPCR) assay using TiLV-93F/TiLV-93R primers and a TiLV93Probe targeting segment 3 of TiLV. The standard curve method was used to determine a PCR amplification efficiency of 92.9% over a wide linear range of 2.7×10^4 to 2.7×10^{10} TiLV copies. The intra-assay and inter-assay coefficients of variation are in the ranges of 0.54–2.18% and 0.59–3.88%, respectively. The TaqMan assay detected TiLV at 2.45×10^5 to 2.45×10^8 copy numbers in liver samples from 17 Nile and red hybrid tilapia samples from 2 countries and showing clinical signs of TiLVD. No fluorescence and thus no TiLV detection was found in liver samples from 11 healthy Nile and red hybrid tilapia. The specificity of the assay was further demonstrated by its inability to detect specimens known to have been infected with common fish pathogens, such as, Iridovirus, *Streptococcus agalactiae*, *Francisella* spp., *Flavobacterium* spp., and *Aeromonas hydrophila*. This newly developed TaqMan RT-qPCR assay can be used as an essential tool for TiLV diagnostics in disease surveillance and control programs, as well as in TiLV basic research projects.

1. Introduction

Tilapia lake virus disease (TiLVD) is an emerging viral disease associated with high mortalities in farm-raised and wild tilapia (Eyngor et al., 2014; Jansen and Mohan, 2017). To date, the etiological agent of this disease, tilapia lake virus (TiLV) has been officially reported on three continents covering twelve countries: Israel (Eyngor et al., 2014), Ecuador (Bacharach et al., 2016a; Del-Pozo et al., 2017; Ferguson et al., 2014), Colombia (Kembou Tsofack et al., 2017), Egypt (Fathi et al., 2017; Nicholson et al., 2017), Thailand (Dong et al., 2017; Surachetpong et al., 2017), Malaysia (Amal et al., 2018), India (Behera et al., 2018), Tanzania, Uganda (Mugimba et al., 2018), Taiwan (OIE, 2017a), the Philippines (OIE, 2017b) and Peru (OIE, 2018a). Undoubtedly, the number of countries reporting the presence of TiLV will continue to increase. In 2015, global tilapia production was 6.4 million tonnes, generating a worldwide trade value of 9.8 billion US dollars (FAO, 2017). The virus is such a serious threat to the global tilapia industry and consequently to the means of support of tilapia farmers and to food security in many parts of the world that urgent warnings have been issued internationally by the FAO (FAO, 2017), WorldFish (Jansen and Mohan, 2017) and the OIE (OIE, 2018b).

To date, only tilapines have shown susceptibility to the disease caused by TiLV, even when the tilapia have been co-farmed with other fish such as different species of carp and mullet (Behera et al., 2018; Eyngor et al., 2014; Fathi et al., 2017). TiLV is most likely proving difficult to control because of the high vulnerability of young tilapines to the disease (Amal et al., 2018; Ferguson et al., 2014; Surachetpong et al., 2017), and due to the fact that there are currently no available TiLV vaccines or anti-viral drugs. This situation is further compounded by a lack of knowledge regarding the pathogenicity of TiLV. Therefore, minimizing the risk of TiLV will depend upon better husbandry and biosecurity measures of which the implementation of rapid, accurate and cost-effective diagnostic tools will be crucial.

TiLV is an enveloped virus consisting of 10 genomic segments of

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linear single-stranded RNA of negative polarity and a total genome size of 10.3 kb (Bacharach et al., 2016a). The virus has been described to be an orthomyxo-like virus due to the fact that all ten segments have similar 5′ and 3′ non-coding regions that are indicative of other viruses in the family of Orthomyxoviridae (Bacharach et al., 2016a). Furthermore, segment 1 of TiLV shows weak sequence homology to the motifs conserved in the influenza C virus polymerase basic 1 (PB1) segment (Bacharach et al., 2016a; Surachetpong et al., 2017). The virus was recently classified by the International Committee on Taxonomy of Viruses (ICTV) as belonging to group V of ssRNA (−), of an undefined family, of the genus *Tilapinevirus* and the species was named *Tilapia tilapinevirus,* where TiLV is a representative isolate (Bacharach et al., 2016b).

Beyond the in vivo examinations and necropsy of infected tilapia, there have been a variety of published laboratory methods used for the detection, isolation, identification and quantification of TiLV (Jansen and Mohan, 2017). TiLV has been successfully isolated and propagated using a variety of cell lines (Behera et al., 2018; Eyngor et al., 2014; Kembou Tsofack et al., 2017; Mugimba et al., 2018; Tattiyapong et al., 2017). Direct demonstration of TiLV virions or nucleic acid has been conducted using electron microscopy (Eyngor et al., 2014; Surachetpong et al., 2017; Tattiyapong et al., 2017) and reverse transcription-PCR based techniques, respectively. Concerning the latter, conventional RT-PCR assays have been established (Eyngor et al., 2014; Mugimba et al., 2018; Nicholson et al., 2017) and to increase sensitivity, various nested and semi-nested RT-PCR assays were developed (Dong et al., 2017; Kembou Tsofack et al., 2017). With the aim of further increasing specificity and sensitivity, a few quantitative RT-PCR (RT-qPCR) methods have been devised using SYBR Green I a non-specific dsDNA binding dye (Kembou Tsofack et al., 2017; Tattiyapong et al., 2018). Typically, the gold standard in virus diagnostics is virus isolation followed by a confirmatory RT-PCR assay. However, despite the strength of cell culture as a primary TiLV diagnostic tool, the method is laborious, time consuming, and not commonly available in routine diagnostic laboratories. Furthermore, conventional endpoint PCR assays require post-PCR processing steps and both methods are often not as sensitive and specific as RT-qPCR approaches. To overcome the limitations of these diagnostic tools, the use of RT-qPCR in the detection of viruses is highly advantageous because of its inherently quantitative nature, high sensitivity, specificity, scalability and its rapid time to result (Bustin, 2000; MacKay, 2004).

We previously established a RT-qPCR assay using SYBR Green I dye and demonstrated that it can accurately quantify TiLV from cell culture and field samples over a wide dynamic range (Liamnimitr et al., 2018; Tattiyapong et al., 2018). We were able to show that it was more sensitive than conventional RT-PCR protocols currently used for TiLV and like others (Hope et al., 2010; Knusel et al., 2007), we were able to show a higher sensitivity over virus isolation (Tattiyapong et al., 2018). However, SYBR green I dye can bind to any amplified product, target or non-target and this cannot be differentiated in the amplification curves. This may lead to false positives and inaccurate quantifications (Bustin and Nolan, 2004; Simpson et al., 2000). In contrast, 5′ nuclease assays, exemplified by TaqMan assays, offer two components for specificity: primers and a probe (Holland et al., 1991). Therefore, in this study, we designed and developed a TaqMan based RT-qPCR assay for the sensitive and specific detection TiLV. According to various validation analyses, we found this new RT-qPCR assay to have high efficiency, sensitivity and specificity and determined it to perform as well as our recently established SYBR green based RT-qPCR assay (Tattiyapong et al., 2018). To maximize the usability of this assay, our primers and probe target an area of TiLV pertaining to the most amount of published sequence information and we were able to demonstrate that the assay detected TiLV not just in Thai tilapia but in tilapia originating from elsewhere too.

2. Materials and methods

2.1. Sample collection and ethical statement

Nile tilapia (*Oreochromis niloticus* L.) and red hybrid tilapia (*Oreochromis* spp.) with clinical signs of TiLV infection ($n = 17$) and healthy fish ($n = 11$) in Thailand and Malaysia were collected during the period of 2015–2018. The fish were euthanized using an overdose of clove oil (Aquanes, Better Pharma, Thailand). The animals used in this study were approved by the Kasetsart Institutional Animal Used Committee under the permit number OAKCU00659 and ACKU 59-VET-016.

2.2. Total RNA extraction, cDNA synthesis, and TiLV segment 3 sequencing

One hundred mg of liver tissue from each field sample was dissected and placed in a sterile 1.5 mL microcentrifuge tube containing 1 mL TRIzol reagent (Thermo Fisher Scientific, USA). In addition to liver, other tissues including brain, spleen, heart, intestine as well as mucus could be used for TiLV detection (Liamnimitr et al., 2018; Tattiyapong et al., 2018). The samples were homogenized and processed for total RNA extraction according to the manufacturer's guidelines. Total RNA concentration was quantified using a Nanodrop 2000 spectrophotometer (Thermo Scientific, USA) and adjusted to 200 ng μL^{-1} using molecular grade water (Wisent Bioproduct, Canada).

For complementary DNA (cDNA) generation, the RNA in each sample was reverse transcribed using ReverTraAce® qPCR RT Master Mix (Toyobo, Japan) containing oligo(dT) and random primers. Briefly, 1 µg of total RNA was mixed with 5× RT Master Mix and the reaction was made up to 10 µL using molecular grade water. The reactions were incubated at 42 °C for 60 min followed by 98 °C for 5 min in a T100 thermal cycler (Bio-Rad Laboratories, USA).

To determine potential nucleotide mismatches at the primers and probe binding sites, RT-PCR assays were performed with cDNA from ten infected fish, being nine from Thailand and one from Malaysia, using Nested ext-2 (5’-TTGCTCTGAGCAAGAGTACC-3′) and Nested ext-1 (5’-TATGCAGTACTTTCCCTGCC-3′) primers amplifying a 491 bp fragment of TiLV genomic segment 3 (Eyngor et al., 2014). The PCR products were purified and Sanger sequenced as previously outlined (Tattiyapong et al., 2018).

2.3. TaqMan assay design and conditions

The nine Thai TiLV sequences and one Malaysian TiLV sequence generated in this study, entitled KUTV11-KUTV20 were deposited in GenBank under accession numbers MH213039-MH213048, respectively. They were aligned with other TiLV partial genomic segment 3 sequences (KU751816.1, MF574205.1, MF582636.1, MF502419.1, and KX631923.1) using the BioEdit sequence alignment editor (Hall, 1999).

TaqMan primers (TiLV-93F 5’-AGCCTGCCACACAGAAG-3′) & TiLV-93R 5’-CTGCTTGAGTTGTGCTTCT-3′) and probe (TiLV-93Probe 5’-FAM-CTCTACCAGCTAGTGCCCCA-Iowa Black-3′) were designed based on the most conserved region of segment 3 of TiLV. The melting temperature, GC contents, and primer-dimer formation were tested by Net Primer [www.premierbiosoft.com/netprimer]. The primers and probe were blasted against the NCBI database to check for sequence similarities in other organisms [www.ncbi.nlm.nih.gov/tools/primer-blast].

The TaqMan qPCR reaction consisted of 0.5 µL Prime PCR assay (Bio-Rad Laboratories, USA), 5 µL iTaq™ universal probes supermix (Bio-Rad Laboratories, USA), 200 ng cDNA template and molecular-grade water to a final volume of 10 µL. The qPCR reactions were performed in triplicate using a CFX96 Touch thermal cycler (Bio-Rad Laboratories, USA). Thermal cycling conditions were an initial activation step at 95 °C for 2 min followed by 40 cycles of 95 °C for 5 s and 56 °C for 30 s. A no-template control (NTC) was included in every instrumental run. Ten-fold serial dilutions of the pTG19-T plasmid

(Vivantis, Malaysia) containing a 491 bp cDNA fragment from TiLV segment 3 (pTiLV) (Tattiyapong et al., 2018) were prepared covering the range 2.7×10^4 to 2.7×10^{10} copies of TiLV. A standard curve was constructed by running RT-qPCR reactions in triplicate and plotting log TiLV copy number versus cycle threshold (Ct). PCR amplification efficiency was calculated as $(100 \times (10^{1/\text{slope}} - 1))$ (Rutledge and Cote, 2003).

2.4. Repeatability, reproducibility, and specificity testing

Repeatability (intra-assay) was calculated from the Ct of technical triplicates of the pTiLV dilution series, while reproducibility (inter-assay) was determined from the mean Ct values of three independent experiments. Both are expressed as coefficients of variation (CV). Assay specificity was examined using specimens from 17 infected and 11 healthy fish. Common bacteria and virus in infected fish tissues, i.e., Iridovirus, *Streptococcus agalactiae*, *Francisella* spp., *Flavobacterium* spp., and *Aeromonas hydrophila* were also tested with the novel TaqMan RT-qPCR assay. Any amplified qPCR products were separated by agarose gel electrophoresis using low melting temperature agarose NuSieve 3:1 agarose (Lonza, Switzerland) and each ethidium bromide stained band was extracted, purified and Sanger sequenced.

3. Results and discussion

3.1. Design of the novel TaqMan RT-qPCR assay based on TiLV segment 3

The aim of this study was to develop and validate a TaqMan probe based RT-qPCR assay for the specific and fast detection of TiLV. We focused on genomic segment 3 of TiLV because as it stands, this region of TiLV has the greatest amount of genomic information publicly available (Behera et al., 2018; Eyngor et al., 2014; Surachetpong et al., 2017). We also augmented this information by providing ten additional genomic segment 3 sequences comprising isolates from Thailand and Malaysia. Specifically, this sequencing information of 10 clinically infected fish were aligned and examined to determine potential nucleotide mismatches at the primers and probe binding sites (Fig. 1). A multiple sequence alignment of segment 3 of the 15 sequences spanning nt 865–967 is presented in Fig. 1. The TiLV sequence isolated from Israel, KU751816.1 (Bacharach et al., 2016a), was set as the reference sequence. The TaqMan primers and probe were designed based on the already published Thai KX631923.1 isolate (Surachetpong et al., 2017), capable of amplifying a 93 nt fragment in the most conserved part of segment 3, specifically, nt 871–963. Using this information, we assessed the amount of primer/probe mismatches across all of the TiLV isolates shown. We found only 1 mismatch in the forward primer, in the probe and in the reverse primer for the Israeli (KU751816.1), Indian (MF574205.1, MF582636.1, and MF502419.1) and Malaysian

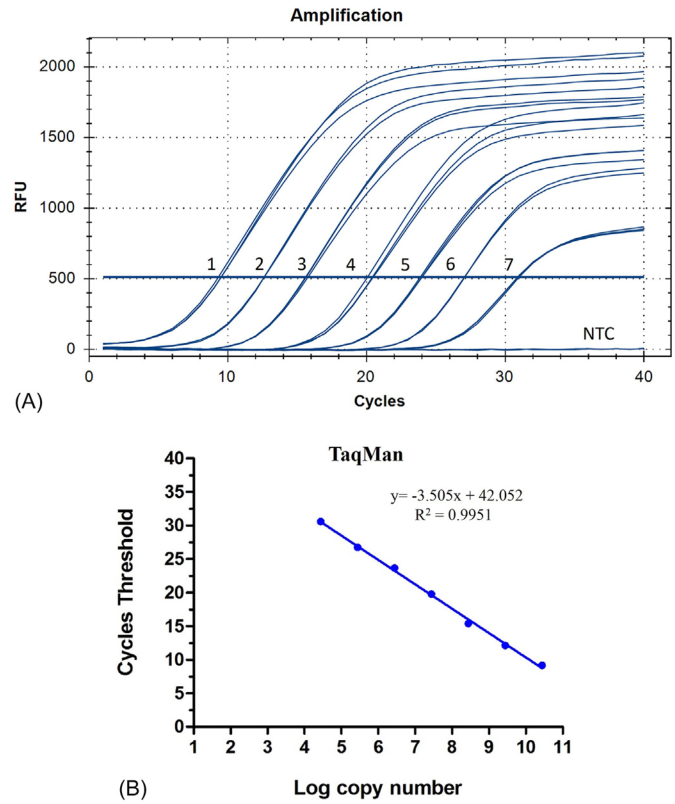

Fig. 2. Amplification curves of TaqMan probe-based RT-qPCR (A) Seven ten-fold serial dilutions of the pTiLV standard plasmid were tested in triplicate. Dilutions labelled 1–7 represent 2.7×10^{10}, 2.7×10^9, 2.7×10^8, 2.7×10^7, 2.7×10^6, 2.7×10^5, 2.7×10^4 copies, respectively and NTC denotes a no template control. (B) Standard curve of RT-qPCR amplification of serially diluted plasmid pTiLV containing segment 3 of TiLV.

(MH213048) isolates. Moreover, none of these mismatches were in critical areas of the primers or probe such as at the extremities. With regards to the Thai strains (MH213039- MH213047), one isolate showed no mismatches, six have 1 mismatch and two have 2 mismatches in the forward primer. We observed one of the Thai isolates, (MH213046) with 2 mismatches in the probe, three isolates have 1 mismatch in the probe and the remaining five showed no nucleotide mismatch in the probe. For binding to the reverse primer, we found one isolate with no mismatches, five with 1 mismatch and three with 2 mismatches. With the exception of one nucleotide mismatch concerning the MH213046 isolate, none of the other observed mismatches are at the extremities of the primers or probe. Definitely, mismatches between

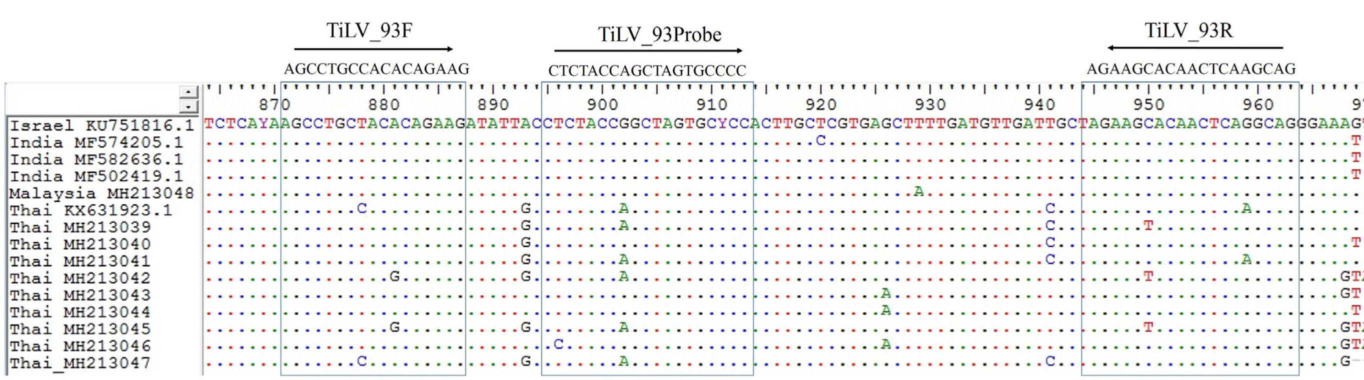

Fig. 1. Partial multiple sequence alignment of segment 3 of tilapia lake virus (TiLV). The orientation and position of the primers and TaqMan probe used in this study are depicted. TiLV isolates from Israel (KU751816.1), India (MF574205.1, MF582636.1 and MF502419.1) and Thailand (KX631923.1) were retrieved from GenBank and the other sequences (Thai isolates: MH213039 – MH213047 and Malaysia isolate: MH213048) were generated in this study.

Table 1
Repeatability and reproducibility of the TaqMan based RT-qPCR using the 10-fold serial dilution of the pTiLV plasmid containing segment 3 of TiLV in three independent experiments.

	Copy number	Intra-assay			Inter-assay		
		Mean Ct	SD	CV (%)	Mean Ct	SD	CV (%)
10^{-1}	2.7×10^{10}	9.48	0.21	2.18	9.40	0.17	1.84
10^{-2}	2.7×10^{9}	12.56	0.20	1.58	12.24	0.48	3.88
10^{-3}	2.7×10^{8}	14.86	0.21	1.39	15.22	0.14	0.90
10^{-4}	2.7×10^{7}	19.65	0.12	0.59	19.84	0.19	0.95
10^{-5}	2.7×10^{6}	23.43	0.13	0.54	23.47	0.14	0.59
10^{-6}	2.7×10^{5}	26.83	0.23	0.84	26.75	0.21	0.80
10^{-7}	2.7×10^{4}	30.76	0.41	1.33	30.67	0.23	0.76

SD: standard deviation.
CV: coefficient of variation.

primers or probes may lead to decreases in sensitivity or even assay failure. However, it has been shown that RT-PCR is robust enough to tolerate a few mismatches depending on their location (Kim et al., 2006).

3.2. Evaluating the RT-qPCR amplification efficiency and sensitivity

We observed consistent amplification curves over the range of the seven pTiLV dilutions meaning that our optimized TaqMan RT-qPCR assay has a wide linear range from 2.7×10^4 copies to 2.7×10^{10} of TiLV per reaction, while the non-template control (NTC) produced no fluorescence and thus no amplification (Fig. 2A). Although the nucleotide mismatches at the primers and probe binding site of pTiLV may affect the sensitivity of the assay at low plasmid copies (below 10^3 copies), we have confirmed that the TaqMan qPCR protocol can amplify TiLV genome in ten-fold serial dilutions of infected fish tissues to as low as 100 pg of RNA from infected fish tissues (data not shown). A standard curve was generated by plotting log copy number versus cycle threshold for each dilution (Fig. 2B), resulting in a slope of −3.505, which was extrapolated to give an efficiency of 92.89%. The coefficient of determination (R^2) was calculated to be 0.9951, indicating that there was a high correlation between cycle number and dilution factor for each dilution series.

3.3. Determining the reliability and reproducibility of the TaqMan based RT-qPCR assay

The repeatability and reproducibility of the TaqMan RT-qPCR method was evaluated by calculating the intra- and inter-assay variations, respectively (Table 1). The %CV of the intra-assay ranged from 0.54% to 2.18% with SD values from 0.12 to 0.41, indicating that the assays were highly repeatable given that the %CV of mean intra-assay Ct values were very low. Likewise, we also found the assays to have high reproducibility with minimum variation between independent experiments, with %CV of mean inter-assay Ct values ranging from 0.59% to 3.88% and SD of 0.14 to 0.48. Similar inter- and intra-assay variations were calculated using cDNA from TiLV infected fish tissues (data not shown).

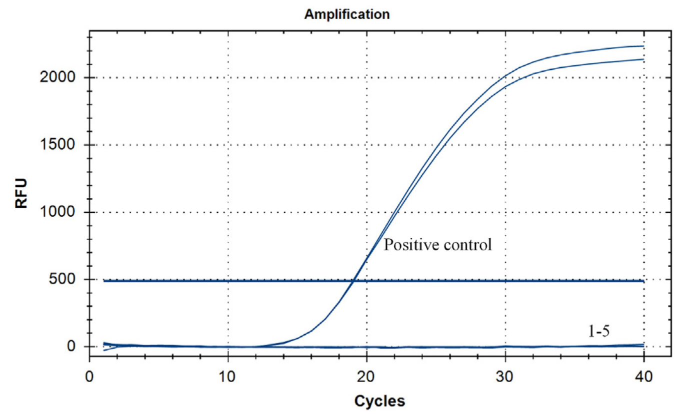

Fig. 3. Specificity of TaqMan-based RT-qPCR for TiLV detection. The amplification curve representing the positive control (pTiLV plasmid) is labelled and samples 1–5 relate to cDNA samples derived from fish tissues infected with Iridovirus, *Streptococcus agalactiae*, *Francisella* spp., *Flavobacterium* spp., and *Aeromonas hydrophila*, respectively.

Table 3
Comparison of the sensitivities and quantitative power of the TaqMan and SYBR Green RT-qPCR assays in detecting TiLV in infected fish tissues.

Fish no.	Ct values (copy number)	
	TaqMan	SYBR Green
1	22.06 (7.85×10^6)	18.36 (7.38×10^6)
2	26.21 (4.47×10^5)	24.39 (2.04×10^5)
3	18.79 (4.90×10^7)	14.90 (5.78×10^7)
4	23.13 (3.10×10^6)	20.36 (2.24×10^6)
5	20.30 (1.86×10^7)	15.47 (4.12×10^7)
6	ND	ND
7	ND	ND
8	ND	ND

ND: no fluorescence detection.

3.4. Assay specificity on field samples and other fish-related pathogens

To test the specificity and applicability of our newly developed assay for use in TiLV detection, we tested TiLV-infected tissues collected from 7 Thai Nile tilapia (*Oreochromis niloticus*), 9 Thai red hybrid tilapia (*Oreochromis* spp.) and 1 Malaysian red hybrid tilapia (Table 2). All 17 infected samples showed amplification curves with Ct values ranging from 16.21–27.14, equating to $2.45 \times 10^{8}-2.45 \times 10^5$ TiLV copy numbers as determined by the standard curve. Indeed, the products generated from positive samples were of the expected 93 bp size and sequence analysis confirmed the presence of expected TiLV sequences (data not shown). In contrast, we observed no amplification (ND) of the cDNA derived from the 11 fish collected with no clinical signs of TiLV infection (Table 2). Furthermore, the specificity of TaqMan RT-qPCR was also tested using cDNA derived from tissue samples infected with

Table 2
Detection of TiLV-infected fish collected from different geographic locations using TaqMan RT-qPCR.

Fish samples	Number of samples	TiLV positive (%)	Ct values (range)	Number of TiLV copies
Positive sample				
Nile tilapia	7	100	16.21–27.14	2.45×10^8–2.45×10^5
Red tilapia	10	100	17.83–24.78	8.91×10^7–1.10×10^6
Negative sample				
Nile tilapia	5	0	ND	ND
Red tilapia	6	0	ND	ND

ND: no fluorescence detection.

five other fish-related pathogens, namely: 1. Iridovirus, 2. *S. agalactiae*, 3. *Francisella* spp., 4. *Flavobacterium* spp., and 5. *A. hydrophila*. No amplification and thus no false positives occurred using such cDNA in our TaqMan RT-qPCR assays (Fig. 3). Collectively, these results show that the newly developed assay was able to specifically amplify the 10 different TiLV isolates coming from Thailand and Malaysia depicted in Fig. 1, meaning that the assay was unaffected by the minor mismatches for each TiLV isolate. Given this result, we can speculate that the assay would also be able to detect the isolate from Israel as well as the Indian isolates.

3.5. Comparison of TaqMan and SYBR green based RT-qPCR assays

We next wanted to compare our novel TaqMan-probe based RT-qPCR assay to our previously established SYBR Green I based RT-PCR assay (Tattiyapong et al., 2018) using 5 positive and 3 negative specimens (Table 3). Comparison of SYBR and TaqMan qPCR protocols revealed that both qPCR methods quantified cDNA prepared from infected materials with equal quantification of viral copy numbers (paired Student's *t*-test; $p > .05$). Despite the fact that TaqMan qPCR is generally considered to be more specific due to the inclusion of the probe, we have found that our TiLV TaqMan and SYBR green RT-qPCR assays perform equally well, as reported by others comparing TaqMan and SYBR green assays (Tajadini et al., 2014).

4. Conclusions

TiLVD is an emerging and highly contagious disease caused by a novel orthomyxo-like virus in tilapia. In this study, we have delivered a sensitive, rapid and specific TaqMan probe-based RT-qPCR assay for the detection of TiLV in field samples. The development of high performance diagnostic tools is of utmost importance for the implementation of screening and containment measures. This novel TaqMan RT-qPCR method can be used for diagnostic purposes or in basic research aiming to better understand TiLV as well as in epidemiological studies of TiLVD.

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